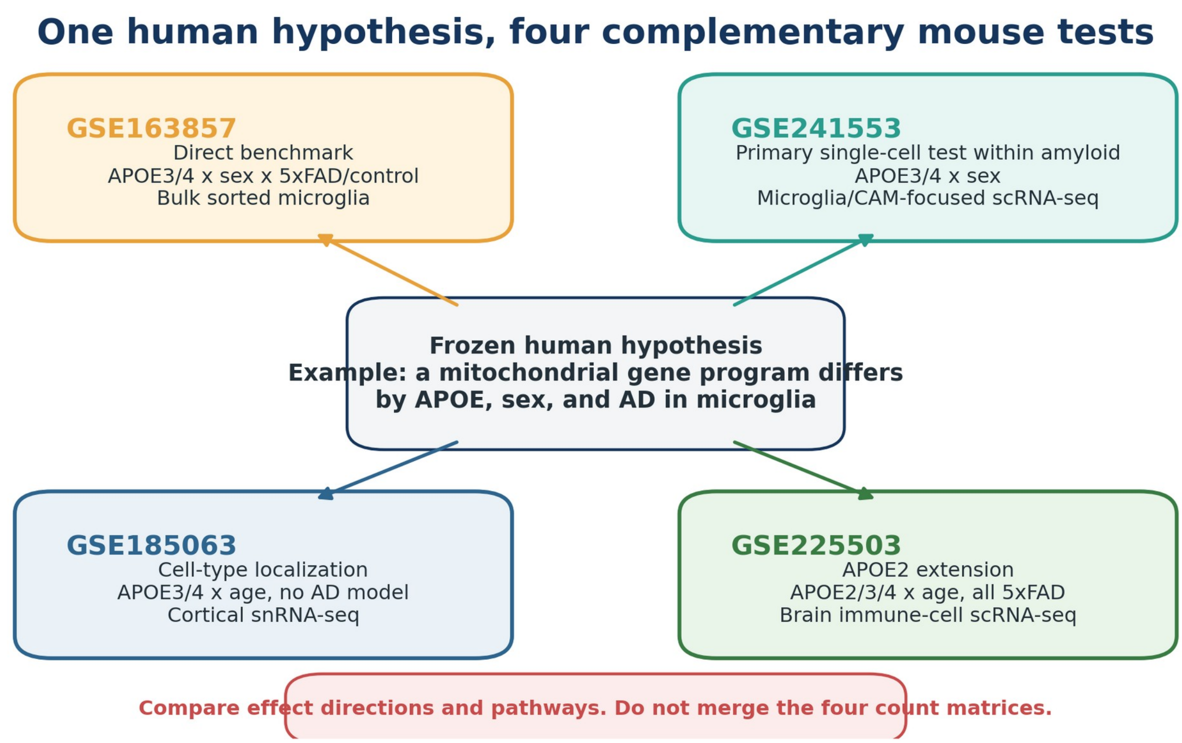
A Beginner's Tutorial for Validating Alzheimer's Disease APOE-by-Sex Findings with Public Mouse RNA-seq Data

A practical, step-by-step guide to GSE163857, GSE241553, GSE185063, and GSE225503



Overview. The four studies answer complementary questions; no single study supplies every factor.

Prepared as a learning guide for a high-school researcher

August 2026

How to use this tutorial

This tutorial assumes that you are new to Alzheimer biology, RNA sequencing, and statistical modeling. It begins with the biological ideas, then explains the statistics, and only then introduces code. You do not need to understand every line of R code on the first reading. First learn what question each step answers; after that, the code becomes much easier to follow.

The most important rule: In single-cell data, thousands of cells from one mouse are still observations from one mouse. For comparisons between experimental groups, the independent biological replicate is normally the mouse, not each cell.

Learning goals

- Explain why four different datasets are needed instead of one perfect dataset.
- Recognize bulk RNA-seq, single-cell RNA-seq, and single-nucleus RNA-seq.
- Build a correct sample metadata table before analyzing gene expression.
- Translate a human gene signature into a fixed mouse gene module.
- Use GSE163857 to test the direct APOE x sex x disease interaction in sorted microglia.
- Use GSE241553 to test APOE x sex within an amyloid-model single-cell experiment.
- Use GSE185063 to ask which cortical cell types carry APOE-related signals.
- Use GSE225503 to extend a question to APOE2 and age while respecting its replication limits.
- Combine conclusions across studies without merging their raw expression matrices.

A sensible order for a beginner

1. Read Parts I-III to learn the biology and statistics vocabulary.
2. Practice with GSE163857 first. Its processed mouse count matrix is small, and every column represents one sample.
3. Move to GSE241553 after you can read a count matrix and metadata table. This introduces single-cell quality control and pseudobulk analysis.
4. Use GSE185063 next to learn cell-type localization in single-nucleus data.
5. Treat GSE225503 as the advanced extension because its processed objects are large and its demultiplexing/replication structure requires careful auditing.
6. Only after the four separate analyses are complete should you make a cross-study evidence table.

Scientific order versus learning order: The learning order starts with the small bulk dataset because it is easier. The scientific report can still present GSE241553 as the primary single-cell result, GSE163857 as the direct factorial benchmark, GSE185063 as broad cell-type localization, and GSE225503 as the APOE2 extension.

Road map

Section	Topic	What you will learn
Part I	Biology starter kit	What AD, APOE, mouse models, and brain cell types mean
Part II	Data and statistics starter kit	Counts, replicates, interactions, modules, orthologs, pseudobulk
Part III	Plan before coding	Freeze the human hypothesis and build the metadata audit
Part IV	Computer setup	R, packages, folders, reproducibility
Part V	GSE163857	Direct APOE x sex x disease benchmark in bulk microglia
Part VI	GSE241553	Within-amyloid APOE x sex single-cell analysis
Part VII	GSE185063	Broad cortical cell-type localization

Section	Topic	What you will learn
Part VIII	GSE225503	APOE2 and age extension in immune cells
Part IX	Evidence synthesis	Compare effect directions without merging studies
Appendices	Templates and glossary	Metadata sheets, exact sample map, code helpers, definitions

Part I. Biology starter kit

1. What is Alzheimer's disease?

Alzheimer's disease (AD) is a progressive brain disorder that damages memory and other thinking abilities. Two famous biological features are amyloid-beta plaques outside cells and abnormal tau protein inside neurons. AD is more complicated than either feature alone: immune responses, blood vessels, lipids, energy metabolism, synapses, and many cell types can all contribute.

Why a mouse model is not the same as a person: A mouse model reproduces selected pieces of human disease. For example, 5xFAD mice develop strong amyloid pathology, but they do not reproduce every feature of late-onset human AD. Use mouse data to test whether a mechanism is conserved, not to claim that the mouse is a complete copy of human disease.

2. What is APOE?

APOE is a gene that encodes apolipoprotein E, a protein involved in moving lipids such as cholesterol. In the brain, APOE is produced mainly by glial cells, including astrocytes and activated microglia. Humans commonly carry APOE2, APOE3, or APOE4. These are called isoforms or alleles: versions of the same gene that differ at a small number of DNA positions.

- APOE3 is the most common human isoform and is often used as the reference group.
- APOE4 is associated with a higher risk of late-onset AD, but carrying APOE4 does not guarantee that a person will develop AD.
- APOE2 is less common and is often associated with lower AD risk, although its biology is not simply the opposite of APOE4.
- A humanized APOE mouse has been genetically engineered so that a human APOE version replaces or supplements the ordinary mouse Apoe context.

Allele versus isoform: An allele is a version of a gene in DNA. An isoform can mean a resulting version of the protein. In this tutorial, APOE2, APOE3, and APOE4 are often called either alleles or isoforms because both ideas are relevant.

3. Why study sex?

Female and male animals can differ in hormones, chromosomes, immune responses, aging, and disease biology. A sex effect asks whether females and males differ on average. A sex-by-APOE interaction asks a more specific question: is the APOE4-versus-APOE3 difference itself different in females versus males? That interaction is often closer to the biological claim in an APOE-by-sex project.

4. Important mouse backgrounds

Term	Plain-language meaning	What it can test	What it cannot prove
5xFAD	A transgenic mouse carrying five familial-AD mutations that produces strong amyloid pathology	Effects in an amyloid-rich AD model	The full biology of common late-onset human AD

Term	Plain-language meaning	What it can test	What it cannot prove
Targeted-replacement control	A mouse carrying human APOE but not the 5xFAD disease transgenes	A matched non-5xFAD comparison for APOE and sex	A completely pathology-free or perfectly human control
Conditional APOE induction	A genetic system in which tamoxifen turns on APOE3 or APOE4 in selected cells	Cell-autonomous consequences of induced APOE expression	AD versus non-AD; tamoxifen control is not a disease control
APOE knock-in, no AD model	Human APOE3 or APOE4 is present without an amyloid/tau transgenic disease model	Baseline or aging-related APOE effects	An APOE-by-AD interaction

5. Brain cell types

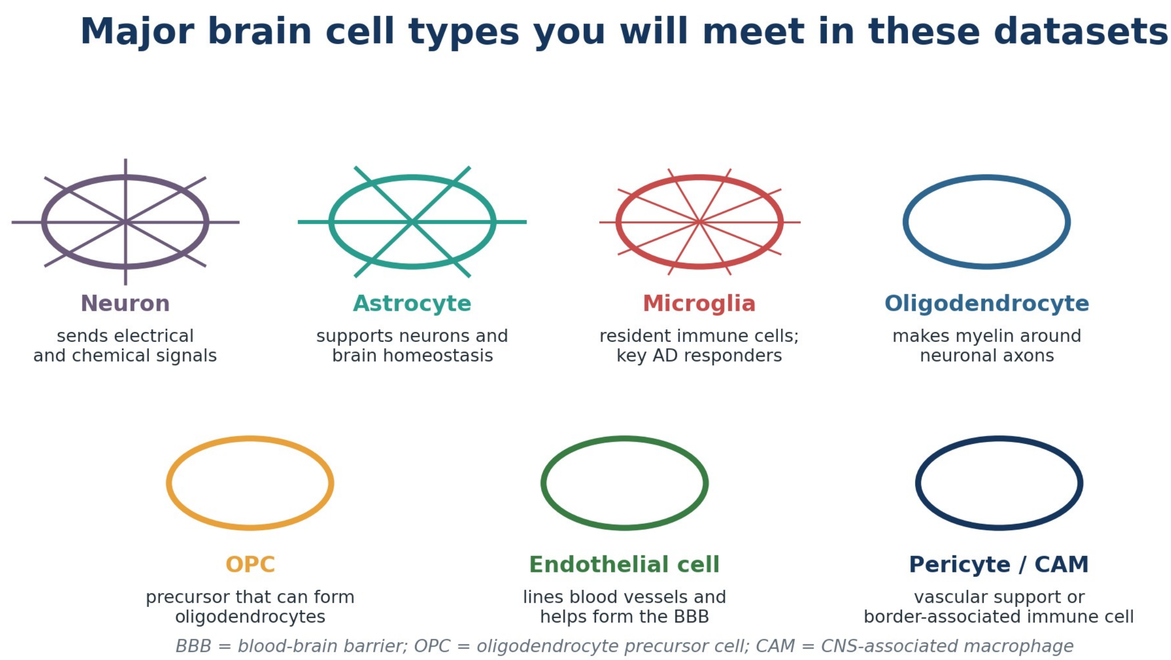


Figure 1. Major cell types encountered in the four datasets. The drawings are schematic.

The same gene program can behave differently in different cell types. A whole-tissue average can hide this. For example, a mitochondrial module may decrease in microglia but stay unchanged in neurons. This is why cell-type localization is one of the most valuable uses of single-cell and single-nucleus data.

6. Brain regions

- Cortex: the outer brain region involved in many higher functions. GSE241553 and GSE185063 use cortical tissue.
- Hippocampus: a region important for memory. GSE225503 includes hippocampal and cortical immune cells.
- Different regions have different cell mixtures and disease timing. A result in cortex is not automatically the same as a result in hippocampus.

Part II. Data and statistics starter kit

7. RNA and gene-expression counts

Genes are DNA instructions. When a cell uses a gene, it produces RNA. RNA sequencing counts fragments of RNA that can be assigned to genes. A count matrix is a table with genes in rows and samples, cells, or nuclei in columns. A larger count usually suggests more RNA from that gene, but library size, technical noise, and cell composition must be handled before comparing groups.

8. Bulk, single-cell, and single-nucleus RNA-seq

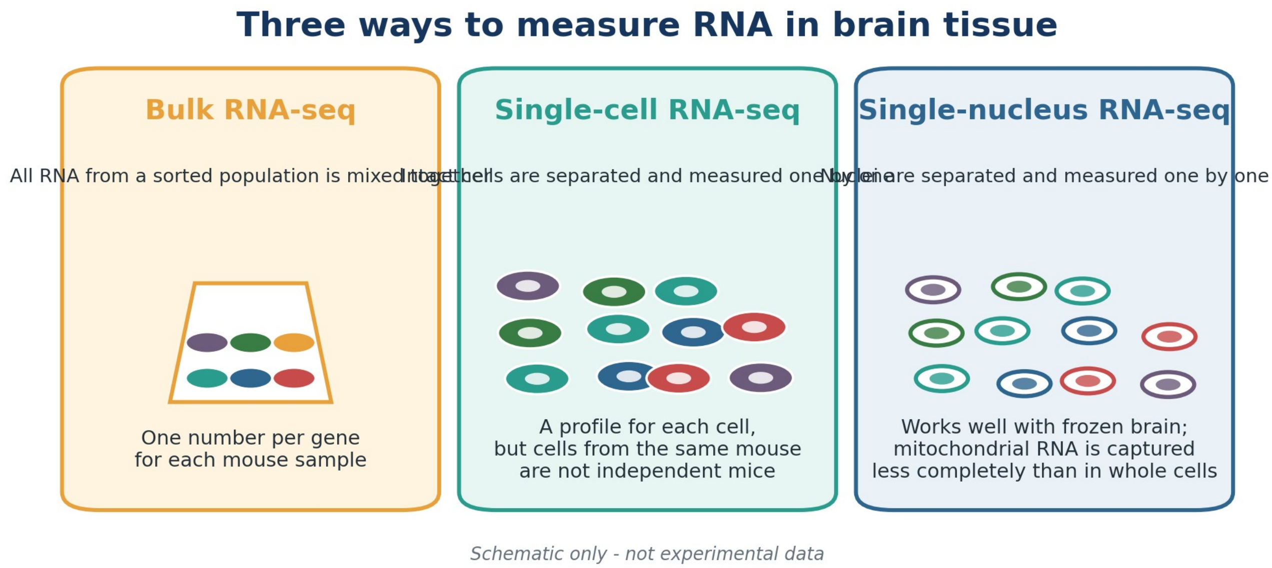


Figure 2. Bulk RNA-seq, scRNA-seq, and snRNA-seq measure related but different units.

Method	What one column means	Strength	Main caution in this project
Bulk RNA-seq	One mouse sample or one sorted-cell population from a mouse	Simple replication and strong counts	No information about variation among individual cells
scRNA-seq	One cell	Cell types and cell states can be separated	Cells from one mouse are nested within that mouse
snRNA-seq	One nucleus	Works well for frozen brain and fragile cell types	Cytoplasmic and mitochondrial transcripts are captured less completely

9. Samples, replicates, and pseudoreplication

A biological replicate is an independently sampled biological unit, such as a mouse. A technical replicate repeats a measurement from the same biological unit. Pseudoreplication happens when non-independent observations are treated as if they were independent. In single-cell experiments, it is incorrect to use thousands of cells from three mice as though there were thousands of mice.

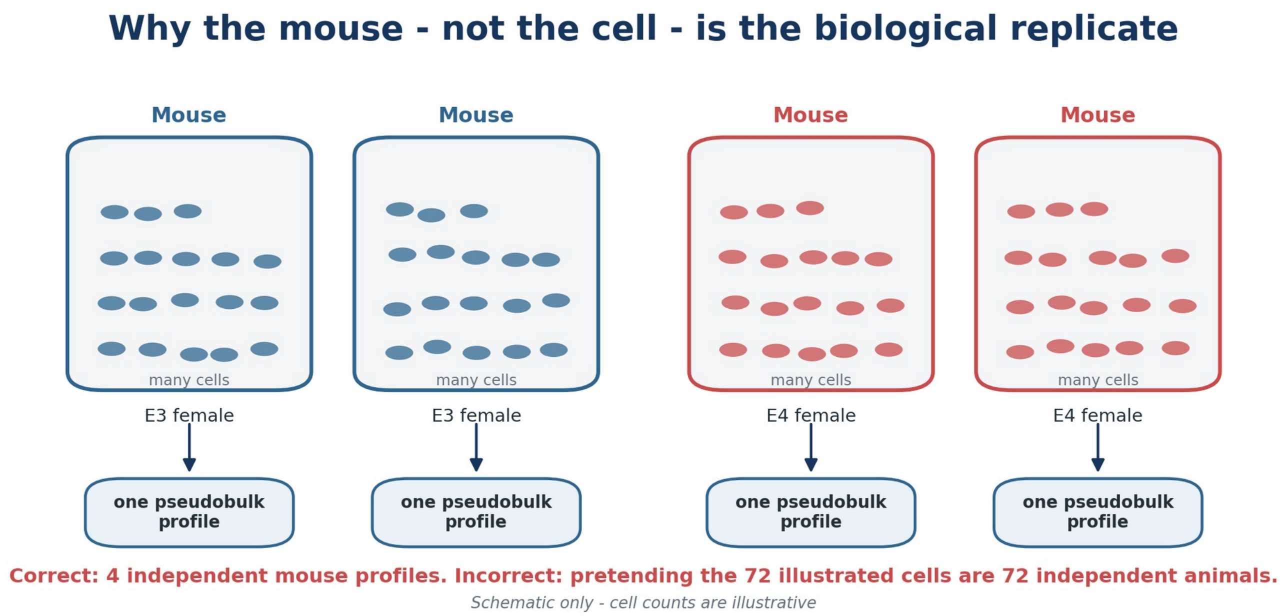


Figure 3. Pseudobulk combines cells of the same type within each mouse, preserving the mouse as the replicate.

Pseudobulk: For each mouse and each cell type, add the raw counts from all matching cells. The result is one gene-count profile per mouse per cell type. This lets a bulk RNA-seq method such as DESeq2 compare mice while still using single-cell labels.

10. Factors, levels, and contrasts

Word	Meaning	Example
Factor	An experimental variable	sex, APOE, disease, age
Level	A category inside a factor	sex has female and male levels
Contrast	A specific comparison	APOE4 minus APOE3 among female 5xFAD mice
Reference level	The baseline used to interpret coefficients	male, APOE3, control
Covariate	A measured variable included to control or explain variation	age or sequencing batch

11. Main effects and interactions

A main effect is an average difference across the other factors in the statistical model. An interaction asks whether one effect changes depending on another factor. For this project, interactions are often more important than main effects.

- sex effect: average female-versus-male difference under the chosen reference conditions.
- APOE effect: average APOE4-versus-APOE3 difference under the chosen reference conditions.
- sex x APOE interaction: whether the APOE4-versus-APOE3 difference changes between females and males.
- sex x APOE x disease interaction: whether the sex-dependent APOE effect changes again between 5xFAD and control backgrounds.

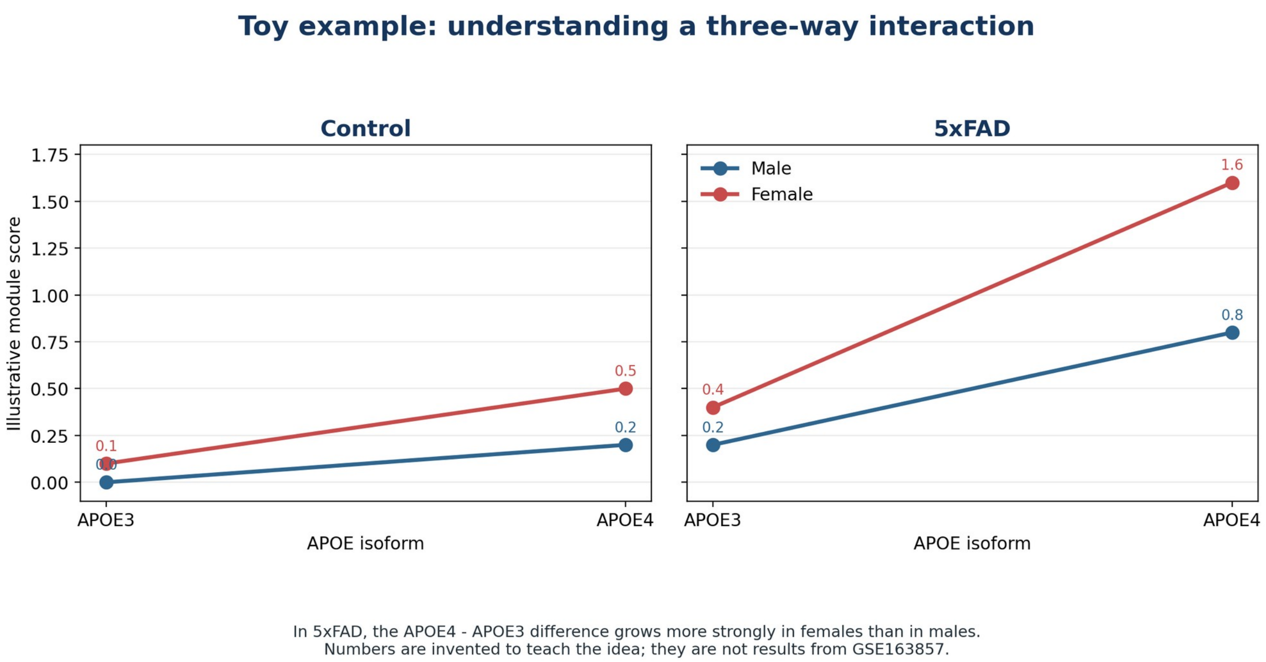


Figure 4. Toy numbers illustrating a three-way interaction. These values are invented for teaching.

Read an interaction as a difference of differences: First calculate APOE4 minus APOE3 in females. Then calculate APOE4 minus APOE3 in males. Subtract those two differences. For a three-way interaction, compare that sex-dependent APOE difference between disease and control backgrounds.

12. Effect size, confidence interval, p value, and FDR

Quantity	Question it answers	How to interpret it
Effect size	How large is the estimated biological difference?	A log2 fold change of +1 means approximately twofold higher expression.
Confidence interval	How uncertain is the estimated effect?	A wide interval means the data allow many plausible effect sizes.
p value	How surprising is the result if the tested effect were truly zero?	It is not the probability that the hypothesis is true.
Adjusted p value / FDR	How is multiple testing controlled across many genes?	Use it when screening thousands of genes; do not ignore effect size.

Small groups require humility: GSE163857 contains all eight factor combinations, but some control cells have only one to three mice. A large estimated interaction with a wide confidence interval is suggestive, not definitive. A non-significant result can mean low power rather than no biology.

13. Gene modules and pathway scores

A gene module is a fixed list of genes representing a biological program, such as oxidative phosphorylation, interferon response, or lipid handling. A module score summarizes the genes into one number per sample or cell. Module-level analysis is often more stable and easier to interpret than testing every gene separately, especially with small sample sizes.

Frozen module: Choose and map the gene list before looking at the mouse outcomes. Do not repeatedly add or remove genes until the mouse result looks significant. That would make the validation circular.

14. Human-to-mouse orthologs

An ortholog is a gene in another species that descended from the same ancestral gene and often has a related function. Human and mouse gene names are not always identical. A one-to-one ortholog map keeps genes for which one human gene matches one mouse gene. This loses some genes, but it reduces ambiguity and makes the validation easier to defend.

15. Batch and confounding

A batch is a technical group, such as a sequencing run or processing day. A confounder is a variable that is mixed up with the experimental factor. For example, if all APOE3 samples were processed in January and all APOE4 samples in June, APOE and batch would be inseparable. Metadata auditing is therefore a scientific step, not paperwork.

Part III. Plan the validation before writing code

16. State one precise human hypothesis

Write the human finding as a sentence with a cell type, gene program, comparison, direction, and brain region. Avoid a vague claim such as "APOE4 changes mitochondria."

> **Worked example - illustrative only:** In human prefrontal-cortex microglia, an oxidative-phosphorylation gene module is lower in female APOE4 AD donors than in female APOE3 AD donors, and this APOE4-versus-APOE3 decrease is larger in females than in males.

This sentence creates testable pieces. The cell type is microglia. The module is fixed. The key contrast is APOE4 minus APOE3. The interaction asks whether that contrast differs by sex. The disease context is AD. A mouse study can support some or all of these pieces, but only if its design contains the necessary factors.

17. Translate the claim into four mouse questions

Dataset	Question for the example	A supportive result would look like	What not to claim
GSE163857	Does sex modify the APOE4 effect differently in 5xFAD versus control microglia?	The three-way interaction or targeted module contrast has the expected direction.	Cell-type localization beyond sorted microglia.
GSE241553	Within amyloid-model mice, does the APOE4 effect differ by sex in microglia/CAMs?	A sex x APOE interaction in induced mice has the expected direction.	Tamoxifen control equals non-AD control.
GSE185063	Which cortical cell types show an APOE4-versus-APOE3 module shift, and does age modify it?	The expected effect appears in microglia or another biologically relevant cell type.	An APOE x AD effect; there is no AD model.
GSE225503	How do APOE2, APOE3, and APOE4 differ across available ages in brain immune cells?	E2/E3/E4 contrasts show a coherent age-dependent pattern.	A sex effect or matched non-AD comparison.

18. Build the metadata audit sheet

Before loading a large expression matrix, create a spreadsheet or CSV with one row per independent mouse or library. Never rely on memory or infer variables from a filename without checking the GEO sample page.

Column	Meaning	Example
dataset	GEO series accession	GSE241553
gsm	GEO sample accession	GSM7730873
library_id	Sequencing library name	MiE-1
mouse_id	Unique animal identifier	MiE-1, if confirmed one mouse/library
sex	Female or male	F
APOE	Human APOE allele/isoform	E3
disease	Disease background	5xFAD, control, or all amyloid
treatment	Induction or treatment status	tamoxifen-induced

Column	Meaning	Example
age	Age in weeks or months	20 weeks
tissue	Region and preparation	cortex; CD45+ cells
cell_type	Assigned later for single-cell data	microglia
batch	Technical processing group	run 1
pooling	Whether multiple mice were combined	yes/no/unknown

Stop condition: Do not run an interaction model until you can draw the factor table and count the independent mice in every cell. If the sex key or mouse identity is unknown, record it as unknown and request metadata.

19. Freeze the human gene module

1. Start from the exact human gene list used in the human study. Keep separate lists for genes expected to increase and genes expected to decrease if your signature has directions.
2. Save the original list with a date and a description of how it was selected.
3. Map human genes to mouse orthologs once. Save the complete mapping table, including genes that failed to map.
4. Keep a one-to-one mouse list for the primary validation. Put ambiguous one-to-many mappings into a sensitivity-analysis list, not the primary list.
5. Do not change the primary list after examining mouse results unless you clearly label the new version exploratory.

R example: freeze a small illustrative mitochondrial module

```
# Example: save a fixed human module
human_module <- c(
  "NDUFS1", "NDUFA9", "COX5A", "ATP5F1A",
  "TOMM20", "TFAM", "PINK1"
)

writeLines(human_module,
           "data/metadata/human_module_frozen_2026-08-22.txt")
```

Important: The seven genes above are only a teaching example. Replace them with the exact pre-specified genes from your human analysis.

20. Map human genes to mouse genes

The Bioconductor package biomaRt can query Ensembl. Because database column names may change, inspect the returned table and save it. A mentor should review the final mapping before publication.

R example: retrieve an auditable human-to-mouse mapping

```
library(biomaRt)

human <- useEnsembl(
  biomart = "genes",
  dataset = "hsapiens_gene_ensembl"
)
mouse <- useEnsembl(
  biomart = "genes",
  dataset = "mmusculus_gene_ensembl"
)

human_ids <- getBM(
  attributes = c("hgnc_symbol", "ensembl_gene_id"),
  filters = "hgnc_symbol",
  values = human_module,
  mart = human
)

hom <- getHomologs(
  human_ids$ensembl_gene_id,
  species_from = "human",
  species_to = "mouse"
)
```

```
names(hom)       # inspect the actual returned columns
head(hom)

# Query mouse symbols for the returned mouse Ensembl IDs.
# Adapt the column name below after inspecting names(hom).
mouse_ids <- unique(hom$mmusculus_homolog_ensembl_gene)
mouse_map <- getBM(
  attributes = c("ensembl_gene_id", "mgi_symbol"),
  filters = "ensembl_gene_id",
  values = mouse_ids,
  mart = mouse
)

write.csv(human_ids, "results/tables/human_ids.csv", row.names = FALSE)
write.csv(hom, "results/tables/human_mouse_homologs_raw.csv", row.names = FALSE)
write.csv(mouse_map, "results/tables/mouse_symbols.csv", row.names = FALSE)
```

Dataset	Modality / tissue	APOE	Sex	Disease background	Replication note
GSE163857	Bulk sorted microglia	APOE3/4	Female/male	5xFAD/control	30 mouse samples
GSE241553	Cortex scRNA-seq	APOE3/4	3F + 3M/group	All amyloid model	24 individual mice
GSE185063	Cortex snRNA-seq	APOE3/4	Sex map missing	No AD model	16 mice; 4/group
GSE225503	CD45+ scRNA-seq	APOE2/3/4	Sex not exposed	All 5xFAD	Age grid is incomplete

Counts and designs are derived from public GEO sample records; always re-audit metadata before analysis.

Figure 5. Public design coverage and reconstructed GSE163857 mouse counts.

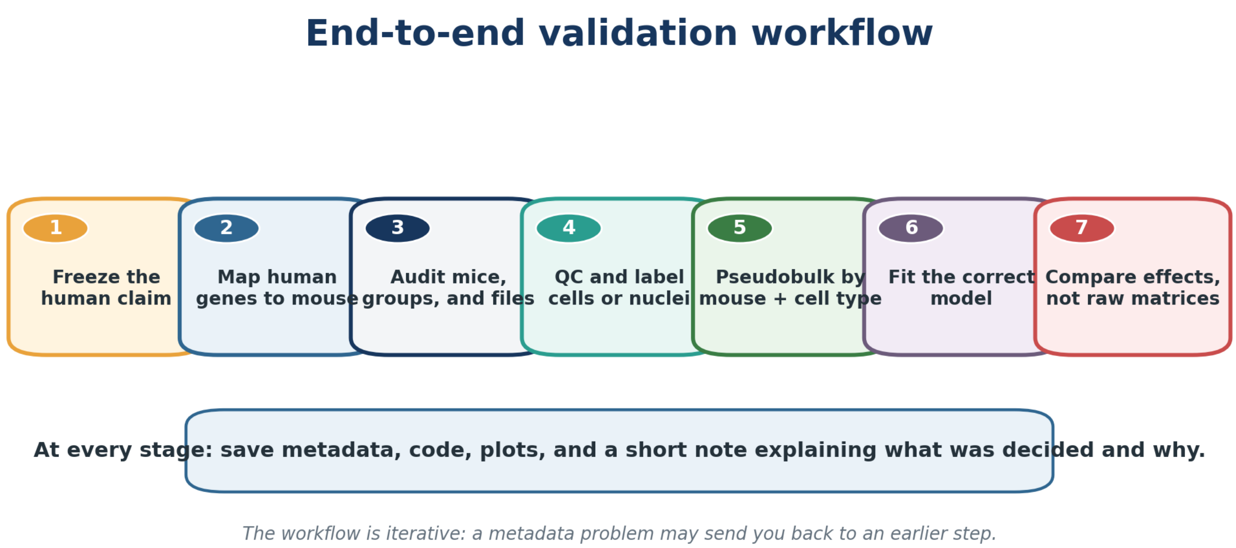


Figure 6. The shared workflow used for all four datasets.

Part IV. Computer setup and project organization

21. Recommended software

The code examples use R because Seurat and DESeq2 are widely used for single-cell and RNA-seq analysis. Install R and RStudio Desktop. The exact package versions will change over time, so record your session information every time you generate final results.

Tool	Purpose	Beginner note
R	Programming language for statistics	Install from the official R project.
RStudio	Editor and interface for R	Makes scripts, plots, and files easier to manage.
Seurat	Single-cell object handling, QC, clustering, pseudobulk	Useful for GSE241553, GSE185063, and GSE225503.
DESeq2	Replicate-aware differential expression from raw counts	Useful for GSE163857 and pseudobulk counts.
GEOquery	Download GEO metadata and supplementary files	Can retrieve files directly from R.
biomaRt	Gene annotation and ortholog mapping	Used before outcome analysis.

R setup: install packages

```
# Install ordinary CRAN packages once
install.packages(c(
  "Seurat", "Matrix", "dplyr", "readr", "stringr",
  "tidyr", "ggplot2", "patchwork", "pheatmap", "R.utils"
))

# Install Bioconductor packages once
if (!requireNamespace("BiocManager", quietly = TRUE)) {
  install.packages("BiocManager")
}
BiocManager::install(c("DESeq2", "GEOquery", "biomaRt"))
```

22. Folder structure

Suggested project layout

```
apoe_mouse_validation/
  data/
    raw/              # downloaded files; never edit them
    metadata/         # sample sheets and frozen gene modules
    processed/        # saved Seurat objects and pseudobulk matrices
```

```
  scripts/
    00_setup.R
    01_ortholog_map.R
    10_gse163857.R
    20_gse241553.R
    30_gse185063.R
    40_gse225503.R
    90_cross_study_summary.R
  results/
    tables/
    figures/
    logs/
  README.md
```

Reproducibility habit: Never manually edit a downloaded count matrix. Write code that reads the original file and produces a new processed file. At the end of every script, save sessionInfo() to a text file.

R example: record package and R versions

```
dir.create("results/logs", recursive = TRUE, showWarnings = FALSE)
writeLines(capture.output(sessionInfo()),
           "results/logs/sessionInfo.txt")
```

23. Computer-resource expectations

Dataset	Public processed size	Practical note
GSE163857	Mouse count CSV about 1.9 MB	Easy on an ordinary laptop; begin here.
GSE241553	Processed MTX/TSV archive about 335 MB	Manageable on a modern computer; merged object may use several GB.
GSE185063	H5 archive about 579 MB	Manageable, but 16 objects and integration can require several GB.
GSE225503	Processed RDS objects about 2.0-2.5 GB compressed	Advanced; 16 GB RAM may be tight, and 32 GB is safer.

Part V. GSE163857 - direct APOE x sex x disease benchmark

24. What this dataset is

GSE163857 contains bulk RNA-seq from fluorescence-activated cell sorting (FACS) of mouse microglia. FACS is a method that labels cells and sorts the desired population. The mice carry human APOE3 or APOE4 on either a targeted-replacement control background or a 5xFAD background, and both sexes are represented. Therefore, this is the one dataset in the four-study plan that directly contains APOE, sex, and disease background together.

Why start here: The processed mouse matrix is small, and each mouse sample is already one column. You can learn metadata, raw counts, model formulas, contrasts, module scores, and interaction interpretation before dealing with millions of cell-level values.

25. The eight experimental groups

Sex	APOE	Disease background	Mouse samples
Female	APOE3	Control/TR	2
Female	APOE3	5xFAD	5
Female	APOE4	Control/TR	2
Female	APOE4	5xFAD	5
Male	APOE3	Control/TR	1
Male	APOE3	5xFAD	7
Male	APOE4	Control/TR	3
Male	APOE4	5xFAD	5

Interpretation warning: The male APOE3 control cell has only one mouse, and several other control cells have two or three. The full model can be fit, but the uncertainty may be large. Report group counts next to every important effect.

26. Download the processed mouse matrix

On the GEO page for GSE163857, download GSE163857_Mouse_Microglia_counts.csv.gz from the supplementary files. The same file can also be downloaded with GEOquery. Keep the .gz file unchanged in data/raw/GSE163857/.

R example: download GSE163857 supplementary files

```
library(GEOquery)

dir.create("data/raw/GSE163857", recursive = TRUE, showWarnings = FALSE)
getGEOSuppFiles(
  "GSE163857",
  makeDirectory = FALSE,
  baseDir = "data/raw/GSE163857"
)

list.files("data/raw/GSE163857", full.names = TRUE)
```

27. Read counts and construct metadata

The public mouse sample names contain sex, APOE, and disease labels. Parsing filenames is convenient, but compare the result against the GEO sample list. The check that metadata rows and count-matrix columns match exactly is essential.

R example: reconstruct and verify the eight groups

```
library(DESeq2)
library(stringr)

count_file <- "data/raw/GSE163857/GSE163857_Mouse_Microglia_counts.csv.gz"
counts <- read.csv(
  gzfile(count_file),
  row.names = 1,
  check.names = FALSE
)
counts <- round(as.matrix(counts))

sample_names <- colnames(counts)
meta <- data.frame(
  sample_id = sample_names,
  sex = ifelse(str_detect(sample_names, "^F"), "F", "M"),
  APOE = ifelse(str_detect(sample_names, "_E4/E4_"), "E4", "E3"),
  disease = ifelse(str_detect(sample_names, "_FAD_"), "FAD", "TR")
)
rownames(meta) <- meta$sample_id

# Put count columns in exactly the same order as metadata rows.
counts <- counts[, rownames(meta), drop = FALSE]
stopifnot(identical(colnames(counts), rownames(meta)))

# The table should match the eight-group counts shown above.
with(meta, table(sex, APOE, disease))
write.csv(meta, "data/metadata/GSE163857_sample_metadata.csv",
          row.names = FALSE)
```

28. First quality checks

1. Check that counts are non-negative integers.
2. Calculate each sample's library size by summing all gene counts.
3. Make a bar plot of library size and label every sample.
4. Create a variance-stabilized PCA plot colored by disease, shaped by sex, and labeled by APOE.
5. Investigate an outlying sample before removing it. Removal needs a documented technical or quality reason, not merely an inconvenient result.

R example: basic quality checks

```
# Basic count checks
stopifnot(all(counts >= 0))
stopifnot(all(counts == round(counts)))

library_sizes <- colSums(counts)
barplot(library_sizes,
        las = 2,
        ylab = "Total raw counts",
        main = "GSE163857 library sizes")
```

29. Fit the full factorial model

DESeq2 uses raw integer counts and a statistical model designed for RNA-seq count variation. Set reference levels first so that coefficients have a clear meaning. The formula sex * APOE * disease expands to all main effects, all two-way interactions, and the three-way interaction.

R example: direct factorial model

```
meta$sex <- relevel(factor(meta$sex), ref = "M")
meta$APOE <- relevel(factor(meta$APOE), ref = "E3")
meta$disease <- relevel(factor(meta$disease), ref = "TR")

# Remove very low-count genes before modeling.
keep <- rowSums(counts >= 10) >= 3
counts_f <- counts[keep, , drop = FALSE]

dds <- DESeqDataSetFromMatrix(
  countData = counts_f,
  colData = meta,
  design = ~ sex * APOE * disease
)
dds <- DESeq(dds)

resultsNames(dds)  # print and save the actual coefficient names
writeLines(resultsNames(dds),
           "results/logs/GSE163857_resultsNames.txt")
```

Do not guess coefficient names: Run resultsNames(dds) and copy the exact coefficient name produced by your installed DESeq2 version. Reference levels and naming conventions determine how the interaction is written.

30. Interpret the coefficients step by step

Coefficient type	Plain-language meaning under M / E3 / control references
sex_F_vs_M	Female minus male among APOE3 control mice.
APOE_E4_vs_E3	APOE4 minus APOE3 among male control mice.
disease_FAD_vs_TR	5xFAD minus control among male APOE3 mice.
sex:APOE	How the APOE4 effect differs in females versus males on the control background.
APOE:disease	How the APOE4 effect changes in 5xFAD versus control among males.
sex:APOE:disease	How the sex-dependent APOE effect changes in 5xFAD versus control.

For a targeted human claim, the most informative quantity may not be a single default coefficient. You may want a contrast such as female APOE4 5xFAD minus female APOE3 5xFAD, or the sex difference in that APOE effect. A statistician or experienced bioinformatician should verify the contrast vector.

31. Beginner-friendly combined-group analysis

A second way to think is to combine sex, APOE, and disease into one eight-level group. This is often easier for pairwise comparisons. It does not replace understanding the interaction, but it makes specific contrasts explicit.

R example: targeted pairwise contrasts

```
meta$group <- factor(paste(meta$sex, meta$APOE, meta$disease, sep = "_"))
```

```
dds_group <- DESeqDataSetFromMatrix(
  countData = counts_f,
  colData = meta,
  design = ~ group
)
dds_group <- DESeq(dds_group)

# Example: APOE4 versus APOE3 among female 5xFAD mice
res_F_FAD_E4_vs_E3 <- results(
  dds_group,
  contrast = c("group", "F_E4_FAD", "F_E3_FAD")
)

# Example: APOE4 versus APOE3 among male 5xFAD mice
res_M_FAD_E4_vs_E3 <- results(
  dds_group,
  contrast = c("group", "M_E4_FAD", "M_E3_FAD")
)
```

Do not compare two p values to test an interaction: One contrast being significant and another not significant does not prove that the two effects differ. The interaction must be tested directly in a single model or contrast.

32. Analyze the frozen module

For a small study, a pre-specified module can be more stable than a genome-wide three-way interaction scan. Use transformed values for visualization and module scoring, but keep raw counts for the DESeq2 gene-level model.

R example: one module score per mouse

```
vsd <- vst(dds, blind = FALSE)
mat <- assay(vsd)

mouse_module <- scan(
  "data/metadata/mouse_module_one_to_one.txt",
  what = "character",
  quiet = TRUE
)
module_genes <- intersect(mouse_module, rownames(mat))

# Standardize each gene across mice, then average genes.
z <- t(scale(t(mat[module_genes, , drop = FALSE])))
meta$module_score <- colMeans(z, na.rm = TRUE)

fit_module <- lm(module_score ~ sex * APOE * disease, data = meta)
summary(fit_module)
confint(fit_module)

write.csv(meta,
          "results/tables/GSE163857_module_scores.csv",
          row.names = FALSE)
```

Directional signatures: If the human signature contains an up-gene list and a down-gene list, a simple score is mean(z scores of up genes) minus mean(z scores of down genes). Document the formula and never change it between datasets.

33. Minimum figures for GSE163857

- Eight-group dot plot: one dot per mouse, with group mean and confidence interval. Never show only bars.
- Interaction plot: APOE3 and APOE4 on the x-axis, separate female and male lines, with separate panels for control and 5xFAD.
- Forest plot: module and selected-gene effect estimates with confidence intervals.
- PCA plot: quality-control visualization, not proof of the biological hypothesis.
- Replicate-count table: place it beside the main result so readers can see the imbalance.

34. What a careful conclusion sounds like

Strong wording: The pre-specified microglial module showed an APOE-by-sex pattern in the 5xFAD background that was directionally consistent with the human finding, while the estimated three-way interaction was imprecise because several control strata contained few mice.

Overclaim to avoid: GSE163857 proves that APOE4 causes the human sex-specific mitochondrial phenotype in all brain cell types.

Part VI. GSE241553 - primary within-amyloid single-cell test

35. What this dataset is

GSE241553 contains cortex single-cell RNA-seq from 24 amyloid-model mice. APOE3 or APOE4 is conditionally expressed in microglia and CNS-associated macrophages (CAMs). The public design reports six mice per major group, with three female and three male mice in each group. This balance makes it the strongest public single-cell near-match for APOE-by-sex within an amyloid context.

Critical design fact: Every group is on an amyloid-model background. Control versus tamoxifen-induced means APOE induction status, not AD versus non-AD. Never label the control mice as non-AD controls.

36. What question to test

For the primary single-cell validation, focus on the tamoxifen-induced mice. Among these 12 mice, test whether the APOE4-versus-APOE3 module difference changes by sex. The model is module score ~ sex * APOE. Because each sex-by-APOE cell contains only three mice, emphasize the pre-specified module, effect sizes, confidence intervals, and consistency with GSE163857.

Subset	Primary question	Model	Biological n
Tamoxifen-induced microglia/CAMs	Does APOE4 versus APOE3 differ by sex within amyloid?	sex * APOE	3 mice per sex-by-APOE cell
All 24 mice, optional audit	Does induction interact with APOE and sex?	induction * sex * APOE	Small 3-way cells; exploratory
Other cell types	Are there downstream non-cell-autonomous patterns?	sex * APOE within induced mice	Interpret as secondary effects

37. Download and inspect the 24 matrices

The supplementary archive contains matrix, feature, and barcode files for each sample. It is about 335 MB compressed. After extraction, keep the original files together and build a 24-row metadata table from the GEO sample pages. The sample order is not a simple block order, so do not infer groups from MiE numbers alone.

R example: download and extract GSE241553

```
library(GEOquery)

dir.create("data/raw/GSE241553", recursive = TRUE, showWarnings = FALSE)
getGEOSuppFiles(
  "GSE241553",
  makeDirectory = FALSE,
  baseDir = "data/raw/GSE241553"
)

# Extract the downloaded TAR file outside R or with untar().
untar(
  "data/raw/GSE241553/GSE241553_RAW.tar",
  exdir = "data/raw/GSE241553/extracted"
)
```

38. Exact public group-code map

Sample	Group code	Sample	Group code	Sample	Group code	Sample	Group code
MiE-1	E3-C1-F	MiE-2	E3-C2-F	MiE-3	E3-T1-F	MiE-4	E3-T2-F
MiE-5	E3-C1-M	MiE-6	E3-C2-M	MiE-7	E3-T1-M	MiE-8	E3-T2-M
MiE-9	E3-C3-M	MiE-10	E4-C1-M	MiE-11	E4-C2-M	MiE-12	E4-C3-M
MiE-13	E4-C3-F	MiE-14	E4-C1-F	MiE-15	E4-C2-F	MiE-16	E3-C3-F
MiE-17	E3-T3-M	MiE-18	E4-T3-M	MiE-19	E4-T2-M	MiE-20	E4-T1-M
MiE-21	E4-T1-F	MiE-22	E4-T2-F	MiE-23	E4-T3-F	MiE-24	E3-T3-F

Code meaning: E3 or E4 is APOE isoform; C is control/no induction; T is tamoxifen-induced; the number is the replicate; F or M is sex. Confirm each mapping against the GEO sample record when you build the final metadata CSV.

R example: create the 24-row metadata table

```
group_code <- c(
  "E3-C1-F", "E3-C2-F", "E3-T1-F", "E3-T2-F",
  "E3-C1-M", "E3-C2-M", "E3-T1-M", "E3-T2-M",
  "E3-C3-M", "E4-C1-M", "E4-C2-M", "E4-C3-M",
  "E4-C3-F", "E4-C1-F", "E4-C2-F", "E3-C3-F",
  "E3-T3-M", "E4-T3-M", "E4-T2-M", "E4-T1-M",
  "E4-T1-F", "E4-T2-F", "E4-T3-F", "E3-T3-F"
)

meta241 <- data.frame(
  mouse_id = paste0("MiE-", 1:24),
  group_code = group_code
)
meta241 <- tidyr::separate(
  meta241,
  group_code,
  into = c("APOE", "condition_rep", "sex"),
  sep = "-",
  remove = FALSE
)
meta241$induction <- ifelse(
  substr(meta241$condition_rep, 1, 1) == "T",
  "TAM", "Control"
)
write.csv(meta241,
          "data/metadata/GSE241553_sample_metadata.csv",
          row.names = FALSE)
```

39. Read each 10x matrix

A 10x processed dataset usually has three files per sample: a sparse count matrix, a feature/gene table, and a barcode table. Create one Seurat object per mouse and add mouse_id before merging. This prevents sample identity from being lost.

R example: one Seurat object per mouse, then merge

```
library(Seurat)
library(stringr)

raw_dir <- "data/raw/GSE241553/extracted"
mtx_files <- list.files(
  raw_dir,
  pattern = "matrix\\.mtx\\.gz$",
  full.names = TRUE
)

read_one_mouse <- function(mtx_file) {
  stem <- sub("\\.matrix\\.mtx\\.gz$", "", mtx_file)
  mouse_id <- str_extract(basename(stem), "MiE-[0-9]+")

  x <- ReadMtx(
    mtx = mtx_file,
    features = paste0(stem, ".features.tsv.gz"),
    cells = paste0(stem, ".barcodes.tsv.gz"),
    feature.column = 2
  )

  obj <- CreateSeuratObject(
    counts = x,
    project = mouse_id,
```

```
    min.cells = 3,
    min.features = 200
  )
  obj$mouse_id <- mouse_id
  obj
}

objects <- lapply(mtx_files, read_one_mouse)
seu241 <- merge(
  x = objects[[1]],
  y = objects[-1],
  add.cell.ids = sapply(objects, function(x) unique(x$mouse_id))
)
```

Filename patterns may differ: Run list.files() and inspect several names before using the code. Adjust the stem, feature, and barcode patterns to the actual extracted filenames. A robust script fails loudly if a companion file is missing.

40. Attach mouse-level metadata to every cell

R example: propagate sample metadata to cell barcodes

```
# Match each cell's mouse_id to the 24-row sample table.
idx <- match(seu241$mouse_id, meta241$mouse_id)
stopifnot(!any(is.na(idx)))

seu241$APOE <- meta241$APOE[idx]
seu241$sex <- meta241$sex[idx]
seu241$induction <- meta241$induction[idx]

# Verify that every mouse has exactly one metadata combination.
with(seu241[[]], table(mouse_id, APOE))
with(seu241[[]], table(mouse_id, sex))
with(seu241[[]], table(mouse_id, induction))
```

41. Quality control for single-cell data

Quality control (QC) tries to remove empty droplets, severely damaged cells, and doublets while keeping real biological diversity. Do not copy a universal cutoff without looking at each mouse. Plot distributions by mouse because library quality can differ.

- nFeature_RNA: number of genes detected in a cell. Very low values may indicate empty or damaged droplets; very high values may indicate two cells captured together.
- nCount_RNA: total RNA counts in a cell. It is related to nFeature_RNA and cell size.
- percent.mt: fraction of counts from mitochondrial genes. High values can indicate stressed or damaged cells in scRNA-seq, but the useful cutoff depends on the experiment.
- Doublet: two cells captured in one droplet, producing a mixed expression profile.
- QC must be summarized per mouse so one poor sample does not disappear silently.

R example: inspect QC before filtering

```
seu241[["percent.mt"]] <- PercentageFeatureSet(
  seu241,
  pattern = "^mt-"
)

VlnPlot(
  seu241,
  features = c("nFeature_RNA", "nCount_RNA", "percent.mt"),
  group.by = "mouse_id",
  ncol = 1,
  pt.size = 0
)

# Example starting thresholds only - inspect your plots first.
# filtered <- subset(
#   seu241,
#   subset = nFeature_RNA > 200 &
#            nFeature_RNA < 6000 &
#            percent.mt < 15
# )
```

Do not optimize QC for the desired result: Set QC rules based on technical distributions and standard cell-quality reasoning before testing APOE-by-sex effects. Report the number of retained cells per mouse and cell type.

42. Normalize, reduce dimensions, and cluster

Normalization makes cell profiles more comparable for visualization. Principal component analysis (PCA) summarizes major expression patterns. UMAP places similar cells near one another in a two-dimensional map. Clustering groups cells with similar profiles. UMAP coordinates are a visualization, not direct physical distances or statistical evidence.

R example: a basic exploratory Seurat workflow

```
seu241 <- NormalizeData(seu241)
seu241 <- FindVariableFeatures(seu241)
seu241 <- ScaleData(seu241)
seu241 <- RunPCA(seu241, npcs = 30)
seu241 <- FindNeighbors(seu241, dims = 1:20)
seu241 <- FindClusters(seu241, resolution = 0.4)
seu241 <- RunUMAP(seu241, dims = 1:20)

DimPlot(seu241, group.by = "seurat_clusters", label = TRUE)
DimPlot(seu241, group.by = "mouse_id")
```

43. Assign broad cell types

Use marker genes, the original paper, and ideally the authors' annotation. A marker is a gene expressed strongly in a cell type. No marker is perfect, so use several markers together and examine whether the full cluster profile is biologically coherent.

Broad type	Example markers	Plain-language role
Microglia	P2ry12, Tmem119, Cx3cr1	Resident brain immune cells
CAM/macrophage	Mrc1, Lyve1, Pf4	Immune cells at brain borders or vessels
Astrocyte	Aldoc, Aqp4, Slc1a3	Support and homeostasis
Oligodendrocyte	Mbp, Plp1, Mog	Myelin production
Endothelial	Cldn5, Pecam1, Kdr	Blood-vessel lining
Pericyte / smooth muscle	Pdgfrb, Rgs5, Acta2	Vascular support and contraction

Marker lists are starting points: Species, age, disease, and activation can change marker expression. Preserve the original annotations when reliable, and document every relabeling decision.

44. Create pseudobulk profiles

For the primary question, subset the tamoxifen-induced microglia, then sum raw counts within each mouse. The result should have 12 columns, one per induced mouse. You can repeat the procedure for CAMs or other broad cell types if enough mice contain enough cells.

R example: sum microglial counts by mouse

```
# Assume a reviewed cell_type column has been added.
mg_induced <- subset(
  seu241,
  subset = cell_type == "Microglia" & induction == "TAM"
)

pb <- AggregateExpression(
  mg_induced,
  assays = "RNA",
  group.by = "mouse_id",
  return.seurat = FALSE
)$RNA

# Metadata for the 12 pseudobulk columns
pb_meta <- meta241[
  match(colnames(pb), meta241$mouse_id),
```

```
  c("mouse_id", "sex", "APOE", "induction")
]
rownames(pb_meta) <- pb_meta$mouse_id
stopifnot(identical(colnames(pb), rownames(pb_meta)))
```

45. Fit the primary sex x APOE model

R example: replicate-aware microglial pseudobulk model

```
pb_meta$sex <- relevel(factor(pb_meta$sex), ref = "M")
pb_meta$APOE <- relevel(factor(pb_meta$APOE), ref = "E3")

keep <- rowSums(pb >= 10) >= 3

dds241 <- DESeqDataSetFromMatrix(
  countData = round(pb[keep, ]),
  colData = pb_meta,
  design = ~ sex * APOE
)
dds241 <- DESeq(dds241)
resultsNames(dds241)

# Use the printed name for the sex-by-APOE interaction.
```

For the frozen module, calculate one score per mouse and fit module_score ~ sex * APOE. Plot all 12 mouse points. Because n = 3 in each cell, treat the interaction estimate as a focused validation, not a high-powered discovery scan.

46. Optional: cell-state proportions

A cell-state proportion asks what fraction of a mouse's microglia belong to a state, such as homeostatic or activated microglia. Calculate the fraction separately for each mouse. Do not compare raw numbers of cells because sequencing depth differs among mice.

R example: one cell-type proportion per mouse

```
prop <- seu241[[]] |>
  dplyr::count(mouse_id, cell_type) |>
  dplyr::group_by(mouse_id) |>
  dplyr::mutate(proportion = n / sum(n)) |>
  dplyr::ungroup()

write.csv(prop,
          "results/tables/GSE241553_celltype_proportions.csv",
          row.names = FALSE)
```

> **Composition needs careful statistics:** Proportions are constrained to sum to one. A rise in one cell type forces others to fall. Simple plots are useful, but publication-level inference may require a composition-aware method and mentor review.

47. Minimum deliverables for GSE241553

- A 24-row metadata table with exact GEO group codes.
- QC plots and retained cell counts per mouse.
- A UMAP labeled by broad cell type and another colored by mouse to check sample mixing.
- A microglial pseudobulk module interaction plot with one point per induced mouse.
- A table of cell-type-specific effect estimates, confidence intervals, and usable mouse counts.
- A sentence explicitly stating that all mice are on an amyloid-model background.

Part VII. GSE185063 - broad cortical cell-type localization

48. What this dataset is

GSE185063 contains cortical single-nucleus RNA-seq from 16 humanized APOE3 or APOE4 knock-in mice. The study includes young mice (about 2-3 months) and older mice (about 9-12 months), with four mice per APOE-by-age group. It is valuable because it includes neurons, astrocytes, oligodendrocytes, OPCs, microglia, endothelial cells, pericytes, and other cortical populations.

What it cannot answer: There is no 5xFAD or other matched AD-model factor. Use it to localize APOE and age effects across cell types, not to test an APOE-by-AD interaction.

Do not read F as female: Sample labels such as E3F and E4F use F for flox/flox in the mouse genotype. The public GEO record does not provide a reliable sample-level sex key. Do not infer sex from the suffix.

49. Public group structure

Age group	APOE3 mice	APOE4 mice	Total
Young, about 2-3 months	4	4	8
Older, about 9-12 months	4	4	8
Total	8	8	16

50. Request the missing sex map before sex analysis

A sample-level sex map is required to add sex to the model. Ask for a table, not a verbal summary. The table should link every GSM accession to sex, exact age, unique mouse ID, and batch, and confirm that each library represents one mouse.

Email template: Subject: Sample-level sex metadata for GSE185063

Dear Dr. Barisano,

I am analyzing GSE185063 for an educational Alzheimer's disease validation project. Could you please share a sample-level table linking each GSM accession to biological sex, exact age, unique animal ID, and sequencing or processing batch? Could you also confirm that each GSM represents one mouse and indicate the sex balance within each APOE-by-age group? The public labels E3F/E4F appear to refer to flox/flox, so we do not want to infer sex incorrectly.

Thank you for your help.

51. Download and read the 16 H5 matrices

The supplementary archive contains one 10x H5 count file per sample. Build a metadata CSV with the H5 filename, GSM accession, APOE, age group, sex = unknown, and mouse ID. Do not let the filename parser invent sex.

R example: download and list the 16 H5 files

```
library(GEOquery)
library(Seurat)

dir.create("data/raw/GSE185063", recursive = TRUE, showWarnings = FALSE)
getGEOSuppFiles(
  "GSE185063",
  makeDirectory = FALSE,
  baseDir = "data/raw/GSE185063"
)
untar(
  "data/raw/GSE185063/GSE185063_RAW.tar",
  exdir = "data/raw/GSE185063/extracted"
)

h5_files <- list.files(
  "data/raw/GSE185063/extracted",
  pattern = "\\.h5$",
  full.names = TRUE
)
length(h5_files)  # expect 16 after checking the archive
```

R example: keep one-mouse identity while reading nuclei

```
# Build this mapping manually from GEO and verify every row.
meta185 <- read.csv(
  "data/metadata/GSE185063_sample_metadata.csv",
  stringsAsFactors = FALSE
)
# Expected columns: gsm, h5_file, mouse_id, APOE, age_group,
#                   exact_age, sex, batch

read_one_nucleus_sample <- function(row) {
  x <- Read10X_h5(row$h5_file)
  obj <- CreateSeuratObject(
    counts = x,
    project = row$mouse_id,
    min.cells = 3,
    min.features = 200
  )
  obj$mouse_id <- row$mouse_id
  obj$gsm <- row$gsm
  obj$APOE <- row$APOE
  obj$age_group <- row$age_group
  obj$sex <- row$sex       # keep NA or "unknown" until verified
  obj$batch <- row$batch
  obj
}

objects185 <- lapply(seq_len(nrow(meta185)), function(i) {
  read_one_nucleus_sample(meta185[i, ])
})
```

52. QC differences for nuclei

Nuclei contain less cytoplasm than intact cells. They often have fewer detected genes and a much lower mitochondrial RNA fraction. Intronic reads are expected because nuclei contain pre-mRNA. Therefore, scRNA-seq cutoffs should not be copied blindly to snRNA-seq.

- Plot nFeature_RNA and nCount_RNA by mouse before filtering.
- A low percent.mt value is normal in nuclei; it should not be treated as proof of healthy mitochondria.
- The original processing used broad gene-count and mitochondrial filters, but your final thresholds should follow the observed sample distributions and reproduce the authors' logic.
- For mitochondrial biology, emphasize nuclear-encoded mitochondrial pathways. Many mitochondrially encoded RNAs are poorly measured in snRNA-seq.

53. Label broad cortical cell types

After normalization, PCA, clustering, and UMAP, label broad types using the publication's marker panels. Start broad; do not force a fine subtype label when evidence is weak. The most relevant types for many AD/APOE hypotheses are microglia, astrocytes, endothelial cells, pericytes, neurons, oligodendrocytes, and OPCs.

54. Pseudobulk by mouse and cell type

The dataset's special strength is localization. Build one pseudobulk matrix for each cell type. For each matrix, columns are mice and rows are genes. Only analyze a cell type if most or all groups contain enough nuclei from enough mice. Record the usable mice and cell counts.

R example: pseudobulk one cortical cell type at a time

```
# Example after merging and assigning a reviewed cell_type column.
pb185 <- AggregateExpression(
  seu185,
  assays = "RNA",
  group.by = c("mouse_id", "cell_type"),
  return.seurat = FALSE
)$RNA

# For a simpler first analysis, subset one cell type first.
endo <- subset(seu185, subset = cell_type == "Endothelial")
pb_endo <- AggregateExpression(
  endo,
  assays = "RNA",
  group.by = "mouse_id",
```

```
  return.seurat = FALSE
)$RNA
```

55. Model APOE and age first

Until sex is verified, the defensible model is APOE * age_group. This tests whether the APOE4-versus-APOE3 effect changes between young and older mice. After receiving the sex map, first make a group-count table. Only add sex if both sexes are sufficiently represented across APOE and age.

R example: APOE-by-age analysis without guessing sex

```
meta_endo$APOE <- relevel(factor(meta_endo$APOE), ref = "E3")
meta_endo$age_group <- relevel(
  factor(meta_endo$age_group),
  ref = "young"
)

dds_endo <- DESeqDataSetFromMatrix(
  countData = round(pb_endo),
  colData = meta_endo,
  design = ~ APOE * age_group
)
dds_endo <- DESeq(dds_endo)
resultsNames(dds_endo)

# Add sex only after receiving and auditing the mouse-level key.
# A possible model is ~ sex + APOE * age_group,
# or a larger interaction only if group counts support it.
```

56. Make a cell-type localization map

For each broad cell type, calculate the same frozen module score and APOE4-minus-APOE3 effect. Put the estimates in a heatmap or forest plot. A strong result is not merely that one cell type has the smallest p value. Look for effect direction, confidence interval, detected module genes, usable mouse count, and biological relevance.

Cell type	Young APOE4-E3 effect	Older APOE4-E3 effect	Usable mice	Interpretation
Microglia	estimate + CI	estimate + CI	n	Does it match the human cell type?
Astrocytes	estimate + CI	estimate + CI	n	Is the signal glial but not microglia-specific?
Endothelial	estimate + CI	estimate + CI	n	Does APOE affect BBB-related cells?
Pericytes	estimate + CI	estimate + CI	n	Is vascular support involved?
Neurons	estimate + CI	estimate + CI	n	Is the signal absent, opposite, or present?

Part VIII. GSE225503 - APOE2 and age extension

57. What this dataset is

GSE225503 profiles CD45-positive immune cells isolated from cortex and hippocampus of 5xFAD mice carrying human APOE alleles. CD45 is an immune-cell surface marker. The study includes APOE2, APOE3, and APOE4 at selected ages and provides both single-cell RNA-seq and, for part of the study, chromatin-accessibility data. Its main value here is testing whether APOE2, APOE3, and APOE4 show different immune-cell programs across age.

Two major limits: All mice are on a 5xFAD background, so there is no matched non-AD comparison. Public GEO metadata do not expose a clean sex factor for this analysis.

58. The age-by-allele grid is incomplete

Age	APOE2	APOE3	APOE4	Recommended use
10 weeks	Yes	Yes	Yes	Three-allele comparison
20 weeks	Yes	Yes	Yes	Three-allele comparison
60 weeks	No	No	Yes	No allele comparison; descriptive E4 only
96 weeks	No	Yes	Yes	E4 versus E3 comparison

Do not force a rectangular design: A model cannot estimate APOE2 at 96 weeks when no APOE2 sample exists there. Analyze complete subsets such as 10 and 20 weeks separately or together, and treat 96 weeks as an E3-versus-E4 comparison.

59. Start from the processed microglia object

The GEO page provides a full processed RDS object and a microglia-only RDS object. The microglia-only file is still about 2 GB compressed, so it may expand to much more in memory. Use a computer with enough RAM, close other programs, and save intermediate subsets.

R example: load the processed microglia object

```
library(R.utils)
library(Seurat)

# After downloading GSE225503_mglia_only.rds.gz:
gunzip(
  "data/raw/GSE225503/GSE225503_mglia_only.rds.gz",
  destname = "data/raw/GSE225503/GSE225503_mglia_only.rds",
  remove = FALSE,
  overwrite = TRUE
)

mglia225 <- readRDS(
  "data/raw/GSE225503/GSE225503_mglia_only.rds"
)
object.size(mglia225)
```

60. Audit the object metadata before any test

The young samples used hashtag oligonucleotides (HTOs) for demultiplexing. HTOs are short barcode tags attached to cells before pooling so the cells can later be assigned to groups. A tag that distinguishes APOE alleles does not automatically prove that each cell can be assigned to an individual mouse. You need a unique animal identifier for ordinary replicate-aware inference.

R example: search for animal, allele, age, and HTO fields

```
meta_names <- colnames(mglia225[[]])
meta_names

grep(
  "mouse|animal|sample|sex|hto|hash|geno|apoe|age|orig",
  meta_names,
  value = TRUE,
  ignore.case = TRUE
)

# Inspect candidate fields one by one.
# Replace candidate_name with a real metadata column.
table(mglia225[["candidate_name"]][, 1], useNA = "ifany")
```

Decision gate: If you can verify a unique mouse ID for each cell, proceed to mouse-level pseudobulk and proportion testing. If you can identify only pooled allele groups, use the dataset descriptively for module direction and cell-state patterns; do not calculate cell-level p values as if cells were independent mice.

61. Analyze the complete age subsets

1. At 10 weeks, compare APOE2 versus APOE3 and APOE4 versus APOE3.
2. At 20 weeks, repeat the same two contrasts.
3. Optionally model 10 and 20 weeks together with APOE * age if individual mouse IDs and replicate counts are verified.
4. At 96 weeks, compare APOE4 versus APOE3 only.
5. At 60 weeks, describe APOE4 microglial states but do not claim an allele effect.

> **Do not assume a straight E2-to-E3-to-E4 scale:** Test E2 versus E3 and E4 versus E3 as separate biological contrasts. APOE2 may have unique effects rather than merely being a weaker version of APOE3 or the opposite of APOE4.

62. Pseudobulk or descriptive pathway scoring

If individual mice are identifiable, sum raw counts by mouse and microglial state, then fit a replicate-aware model. If they are not identifiable, compute module scores and show distributions or group summaries as descriptive evidence only. The wording of the conclusion must match the replication level.

R example: the analysis depends on verified metadata column names

```
# Pseudocode: adapt the column names after metadata audit.
verified <- subset(
  mglia225,
  subset = age %in% c("10wk", "20wk") &
           APOE %in% c("E2", "E3", "E4")
)

# Only when mouse_id is a verified individual-animal identifier:
pb225 <- AggregateExpression(
  verified,
  assays = "RNA",
  group.by = c("mouse_id", "microglia_state"),
  return.seurat = FALSE
)$RNA
```

63. Cell-state frequencies

This study describes an exhausted-like or terminally inflammatory microglial state. A useful validation asks whether the frequency of a similar state changes with APOE and age. Calculate state frequency per mouse, not per cell, and show every mouse point. A mentor should review the compositional model because cell-state percentages sum to one.

64. Minimum deliverables for GSE225503

- A screenshot or saved table of all candidate metadata fields and their values.
- A written decision explaining whether unique mice can be recovered.
- Separate 10-week and 20-week E2/E3/E4 effect tables.
- A 96-week E4-versus-E3 effect table, labeled as a different comparison.
- No allele comparison at 60 weeks.
- A clear statement that all groups are 5xFAD and that sex is not part of the public analysis.

Part IX. Combine evidence without merging the studies

65. Why merging the matrices is misleading

The four studies were performed in different laboratories, with different mouse constructions, ages, tissues, cell-isolation methods, sequencing protocols, and disease contexts. If you merge their counts, "dataset" is almost perfectly confounded with biological design. A batch-correction method cannot manufacture missing controls or make tamoxifen induction equivalent to 5xFAD versus control.

> **Correct synthesis:** Analyze each study separately. Then compare the sign, size, uncertainty, cell type, and pathway meaning of matched effects.

66. Make a standard result record for every analysis

Field	Example entry
dataset	GSE241553
cell type	Microglia
subset	Tamoxifen-induced amyloid-model mice
contrast	sex x APOE interaction for frozen module
estimate	-0.42 standardized-score units
95% CI	-0.95 to +0.11
adjusted p value	Not primary for pre-specified module / report model p value
independent mice	3 per sex-by-APOE cell
direction vs human	Same / opposite / unclear
limitations	Small n; within-amyloid only

The numbers above are placeholders showing the format. Replace them with actual estimates. A standardized module effect can be compared qualitatively across studies, but avoid pretending that it is numerically identical across different assays and models.

67. Evidence hierarchy

1. Direct, replicate-aware APOE x sex x disease effect in GSE163857, with reasonable uncertainty.
2. Replicate-aware sex x APOE effect in induced GSE241553 microglia within amyloid.
3. Cell-type-specific APOE effect in GSE185063 that localizes the same frozen module to a relevant population.
4. APOE2/E3/E4 and age pattern in GSE225503 when replication can be verified.
5. Descriptive direction in pooled or incompletely replicated data.
6. Visual similarity without a pre-specified contrast - weakest evidence.

68. A traffic-light synthesis

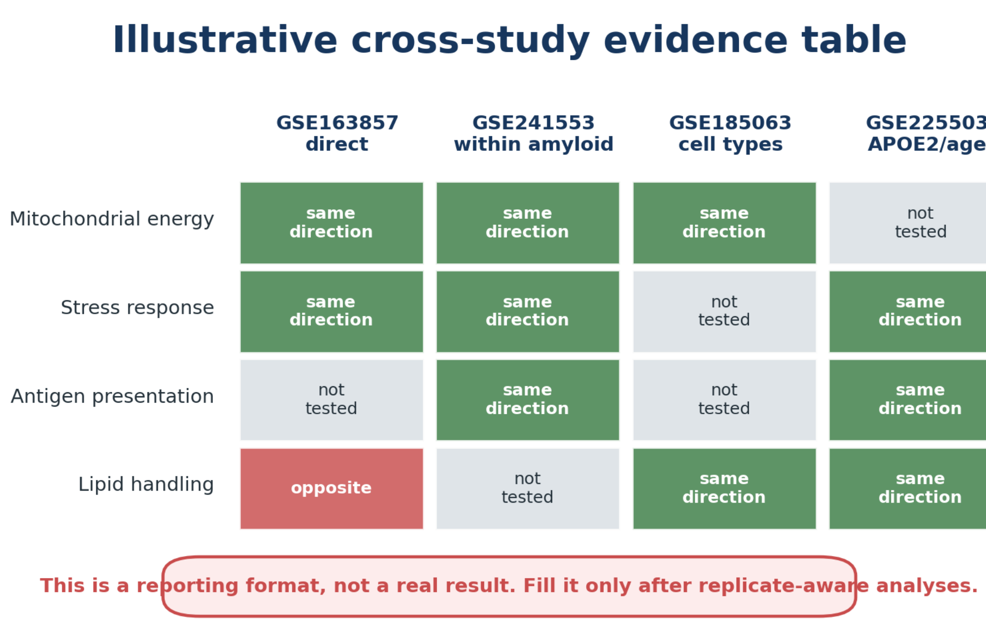


Figure 7. Example format for a cross-study concordance matrix. The shown cells are illustrative, not results.

Color	Meaning	Example wording
Green	Expected direction with replicate-aware support	Consistent support in microglial pseudobulk.
Yellow	Expected direction but wide uncertainty or descriptive replication only	Directionally consistent, but underpowered or pooled.
Red	Clear opposite direction under a comparable contrast	The mouse effect opposed the frozen human prediction.
Gray	The dataset cannot test the claim	No sex factor or no non-AD group.

69. Claim-mapping examples

Final claim	Minimum evidence needed
"The human microglial pathway is conserved in an amyloid mouse context."	Expected module direction in GSE241553 microglia and supportive GSE163857 5xFAD contrast.
"The effect is specific to microglia."	Microglial support plus weaker/absent effects in several adequately powered GSE185063 cell types; absence must have usable data.
"The effect depends on sex."	A direct sex interaction with independent mice; separate male/female significance is not enough.
"The effect depends on AD background."	GSE163857 disease interaction; GSE241553 and GSE225503 alone cannot establish it because all relevant mice are within AD/amyloid backgrounds.
"APOE2 is protective."	A pre-defined E2-versus-E3/E4 direction tied to a biological outcome; expression alone supports mechanism, not clinical protection.

70. Recommended final figure set

- Figure A - dataset design matrix: factors, modality, tissue, independent mice, and major limitations.
- Figure B - GSE163857 interaction plot and forest plot for the frozen module.
- Figure C - GSE241553 microglial UMAP plus mouse-level sex x APOE module plot.
- Figure D - GSE185063 cell-type forest plot or heatmap of APOE effects by age.
- Figure E - GSE225503 E2/E3/E4 age-specific module or state-frequency plot, only at supported ages.
- Figure F - cross-study concordance table with green/yellow/red/gray evidence coding.

71. Suggested Results-section structure

1. Define the frozen human module and one-to-one ortholog mapping.
2. Present the direct factorial microglial benchmark in GSE163857.
3. Present the primary within-amyloid single-cell APOE-by-sex result in GSE241553.
4. Localize APOE effects across cortical cell types with GSE185063.
5. Add APOE2 and age evidence from GSE225503 if the metadata audit supports it.
6. Summarize concordance and disagreements without combining raw matrices.
7. End with limitations: small factor cells, model differences, incomplete sex metadata, pooling, and snRNA mitochondrial capture.

Part X. Common mistakes and troubleshooting

72. Ten mistakes to avoid

Mistake	Why it is wrong	Fix
Treating cells as replicates	Inflates the sample size and produces unrealistically small p values.	Pseudobulk by mouse or use a model with mouse as the experimental unit.
Calling GSE241553 control non-AD	Control refers to induction status; all mice are amyloid-model mice.	Say control/no induction versus tamoxifen-induced.
Reading E3F/E4F as female	F denotes flox/flox in GSE185063 labels.	Request the sample-level sex map.

Mistake	Why it is wrong	Fix
Merging all four matrices	Dataset and design are confounded.	Analyze separately and compare standardized conclusions.
Using normalized counts as DESeq2 input	DESeq2 expects raw or estimated integer-like counts.	Use raw counts; use VST for plots and scores.
Changing the gene module after seeing mouse results	Makes the validation circular.	Freeze the module and mapping first.
Using every age in one complete grid for GSE225503	Some allele-age combinations do not exist.	Analyze supported subsets only.
Claiming no effect from a non-significant result	Small n can create low power and wide intervals.	Report effect size and confidence interval.
Using mitochondrial-gene absence in snRNA as negative evidence	Nuclei capture mitochondrial RNA incompletely.	Emphasize nuclear-encoded mitochondrial pathways.
Ignoring sample order	Counts can be assigned to the wrong metadata row.	Use identical() checks before every model.

73. If a Seurat object is too large

- Start from the authors' processed object instead of raw FASTQ files.
- Use the microglia-only GSE225503 object if your question is microglial.
- Remove assays, reductions, or layers that you are certain you do not need only after saving the original object.
- Process datasets one at a time and save pseudobulk matrices; the final cross-study table is small.
- Use a higher-memory workstation or institutional cluster for GSE225503 rather than weakening the analysis to fit a laptop.

74. If DESeq2 says the model matrix is not full rank

This error means the model asks for effects that the data cannot separate. Common causes include a missing factor combination, a batch perfectly matching a genotype, or including both a combined group and its component factors. Print the group table and simplify the design to effects that the observed data support.

R example: inspect whether the requested design exists

```
# Diagnose the design before fitting.
with(meta, table(sex, APOE, disease))
model.matrix(~ sex * APOE * disease, data = meta)

# For an incomplete age-by-allele design, subset a complete comparison.
meta_10_20 <- subset(meta225, age %in% c("10wk", "20wk"))
with(meta_10_20, table(age, APOE))
```

75. If gene symbols do not match

- Check whether the matrix uses mouse symbols, Ensembl IDs, or symbols with version suffixes.
- Do not silently uppercase mouse symbols to imitate human symbols.
- Save the mapping table and the number of module genes detected in each dataset.
- If fewer than about half of a small module is detected, interpret the module score cautiously and investigate why.

76. If cell types are unclear

- Compare several marker genes, not one marker.
- Read the original paper's annotation methods and supplementary marker tables.
- Check whether the cluster is dominated by one mouse, which can indicate a technical artifact.
- Use broad labels first. "Vascular" may be safer than forcing endothelial versus pericyte when separation is weak.
- Ask an experienced neurobiologist to review disputed labels before publication.

Part XI. Final analysis checklist

[ ] The human hypothesis is written as a precise sentence with cell type, module, direction, sex, APOE, and disease context.
[ ] The human gene module and one-to-one mouse ortholog map are frozen and versioned.
[ ] Every dataset has a saved sample metadata CSV and a group-count table.
[ ] Independent mice, pooled samples, and technical replicates are distinguished.
[ ] GSE163857 counts match the reconstructed eight-cell table.
[ ] GSE241553 control versus tamoxifen is not mislabeled as non-AD versus AD.
[ ] GSE241553 analysis uses mouse-level pseudobulk or mouse-level module scores.
[ ] GSE185063 sex is not inferred from E3F/E4F; the author request is documented.
[ ] GSE185063 primary model is APOE x age until sex is verified.
[ ] GSE225503 metadata fields are audited for unique mouse identifiers.
[ ] GSE225503 age-by-allele comparisons use only observed combinations.
[ ] Raw counts are used for DESeq2; transformed values are used for visualization or module scoring.
[ ] Every important plot shows individual mouse points or clearly states that replication is unavailable.
[ ] Effect sizes and confidence intervals are reported with p values/FDR.
[ ] No raw count matrices from different studies are merged for the final test.
[ ] Cross-study conclusions compare direction, cell type, pathway, and uncertainty.
[ ] All scripts save sessionInfo(), output tables, and figure files.
[ ] A mentor has reviewed model contrasts, metadata ambiguities, and final biological wording.

Appendix A. Reusable metadata templates

A1. Mouse-level sample sheet

CSV template: one row per mouse or library

```
dataset,gsm,library_id,mouse_id,sex,APOE,disease,treatment,age,tissue,batch,pooling,notes
GSE241553,GSM7730873,MiE-1,MiE-1,F,E3,amyloid,Control,reported,cortex,unknown,no,verify from GEO
```

A2. Cell-type analysis audit

dataset	cell_type	group	mice available	cells/nuclei per mouse	included?	reason
GSE241553	Microglia	F-E3-TAM	3	record vector	yes	primary target
GSE185063	Pericyte	old-E4	4	record vector	maybe	exclude if too sparse
GSE225503	TIM-like	10wk-E2	verify	record vector	pending	mouse IDs not yet audited

A3. Cross-study result table

CSV template: one row per pre-specified effect

```
dataset,cell_type,subset,contrast,estimate,ci_low,ci_high,p_value,p_adj,n_mice,direction_vs_human,limitations
GSE163857,Microglia,all mice,sex:APOE:disease,,,,,,,pending,"small control strata"
GSE241553,Microglia,TAM only,sex:APOE,,,,,,,pending,"within amyloid only"
GSE185063,Endothelial,young,APOE4-E3,,,,,,,pending,"no AD model"
GSE225503,Microglia,10 weeks,APOE2-E3,,,,,,,pending,"verify mouse IDs"
```

Appendix B. GSE241553 exact sample-code reference

The following map is transcribed from public GEO sample pages and is included because the MiE order is not grouped cleanly. Use the GSM accession and sample page as the final authority when building metadata.

GSM	Library	Code
GSM7730873	MiE-1	E3-C1-F
GSM7730874	MiE-2	E3-C2-F
GSM7730875	MiE-3	E3-T1-F
GSM7730876	MiE-4	E3-T2-F
GSM7730877	MiE-5	E3-C1-M
GSM7730878	MiE-6	E3-C2-M
GSM7730879	MiE-7	E3-T1-M
GSM7730880	MiE-8	E3-T2-M
GSM7730881	MiE-9	E3-C3-M
GSM7730882	MiE-10	E4-C1-M
GSM7730883	MiE-11	E4-C2-M
GSM7730884	MiE-12	E4-C3-M
GSM7730885	MiE-13	E4-C3-F
GSM7730886	MiE-14	E4-C1-F
GSM7730887	MiE-15	E4-C2-F
GSM7730888	MiE-16	E3-C3-F
GSM7730889	MiE-17	E3-T3-M
GSM7730890	MiE-18	E4-T3-M
GSM7730891	MiE-19	E4-T2-M
GSM7730892	MiE-20	E4-T1-M
GSM7730893	MiE-21	E4-T1-F
GSM7730894	MiE-22	E4-T2-F
GSM7730895	MiE-23	E4-T3-F
GSM7730896	MiE-24	E3-T3-F

Appendix C. Reusable R helpers

C1. Check count-matrix and metadata order

R helper: never let sample order drift

```
check_alignment <- function(counts, meta) {
  if (is.null(rownames(meta))) {
    stop("Metadata needs sample IDs as row names.")
  }
  missing_in_counts <- setdiff(rownames(meta), colnames(counts))
  extra_in_counts <- setdiff(colnames(counts), rownames(meta))
  if (length(missing_in_counts) > 0 || length(extra_in_counts) > 0) {
    stop("Sample IDs differ between counts and metadata.")
  }
  counts <- counts[, rownames(meta), drop = FALSE]
  stopifnot(identical(colnames(counts), rownames(meta)))
  counts
}
```

C2. Simple module score from transformed values

R helper: fixed up-minus-down signature

```
score_module <- function(expression_matrix, genes_up, genes_down = NULL) {
  up <- intersect(genes_up, rownames(expression_matrix))
  if (length(up) < 2) stop("Too few up-module genes were detected.")

  z <- t(scale(t(expression_matrix)))
  up_score <- colMeans(z[up, , drop = FALSE], na.rm = TRUE)

  if (is.null(genes_down)) return(up_score)
```

```
  down <- intersect(genes_down, rownames(expression_matrix))
  if (length(down) < 2) stop("Too few down-module genes were detected.")
  down_score <- colMeans(z[down, , drop = FALSE], na.rm = TRUE)
  up_score - down_score
}
```

C3. Save an auditable result table

R helper: record the comparison before filling in results

```
result_row <- data.frame(
  dataset = "GSE241553",
  cell_type = "Microglia",
  subset = "TAM induced",
  contrast = "sex x APOE",
  estimate = NA_real_,
  ci_low = NA_real_,
  ci_high = NA_real_,
  p_value = NA_real_,
  p_adj = NA_real_,
  n_mice = 12,
  direction_vs_human = "pending",
  limitations = "3 mice per sex-by-APOE cell; all amyloid"
)

write.csv(result_row,
          "results/tables/GSE241553_primary_result.csv",
          row.names = FALSE)
```

Appendix D. Glossary

Adjusted p value / FDR: A p value corrected for testing many genes, designed to control the expected fraction of false discoveries.

Allele: A version of a gene, such as APOE2, APOE3, or APOE4.

Amyloid-beta: A peptide that can accumulate into plaques in Alzheimer pathology.

Batch: A technical processing group, such as a sequencing run or preparation day.

Blood-brain barrier (BBB): Specialized blood vessels and surrounding cells that regulate movement between blood and brain.

Bulk RNA-seq: RNA sequencing that averages a whole tissue or sorted cell population.

CAM: CNS-associated macrophage, an immune cell associated with brain borders or vessels.

Cell state: A functional condition within a cell type, such as homeostatic or inflammatory microglia.

Cell type: A broad biological class, such as microglia, astrocyte, or neuron.

Confidence interval: A range showing uncertainty around an estimated effect.

Confounder: A variable that is entangled with the factor of interest, making effects hard to separate.

Contrast: A specific comparison between groups or combinations of model coefficients.

Count matrix: A table of gene counts by samples, cells, or nuclei.

Differential expression: A statistical comparison asking whether gene expression systematically differs between groups.

Doublet: Two cells captured together and mistakenly measured as one single-cell droplet.

Effect size: The estimated magnitude and direction of a difference.

Factorial design: An experiment containing combinations of two or more factors, such as sex, APOE, and disease.

FACS: Fluorescence-activated cell sorting, a method for isolating cells using fluorescent labels.

Flox/flox: A genotype in which a DNA segment is flanked by loxP sites for conditional genetic manipulation; it does not mean female.

Gene module: A fixed group of genes representing a biological program.

GEO: NCBI Gene Expression Omnibus, a public repository for gene-expression studies.

HTO: Hashtag oligonucleotide, a barcode used to demultiplex pooled single-cell samples.

Interaction: A statistical effect showing that one factor's effect changes depending on another factor.

Isoform: A version of a protein or gene product; APOE2/E3/E4 are commonly called isoforms.

Library: The prepared sequencing material from a biological sample or pooled material.

Log2 fold change: A difference on a base-2 logarithm scale; +1 is about 2-fold, -1 is about half.

Marker gene: A gene used with other evidence to identify a cell type or state.

Metadata: Information describing samples, such as mouse ID, sex, genotype, age, and batch.

Microglia: Resident immune cells of the brain.

Module score: One number summarizing expression of a fixed gene module.

Ortholog: A related gene in another species descended from a common ancestral gene.

PCA: Principal component analysis, a method that summarizes major sources of variation.

Pseudobulk: Raw counts summed across cells of the same type within each biological replicate.

Pseudoreplication: Treating non-independent observations as independent replicates.

Quality control (QC): Checks and filters intended to identify low-quality samples, cells, or nuclei.

Raw count: The unnormalized number of sequencing fragments assigned to a gene.

Reference level: The baseline category used to interpret model coefficients.

Replicate: An independent biological unit, usually an individual mouse in this tutorial.

scRNA-seq: Single-cell RNA sequencing of intact cells.

snRNA-seq: Single-nucleus RNA sequencing of isolated nuclei.

Tamoxifen induction: Use of tamoxifen to activate a conditional genetic system.

UMAP: A two-dimensional visualization that places transcriptionally similar cells near one another.

Variance-stabilizing transformation (VST): A DESeq2 transformation useful for visualization and sample comparison, not raw-count inference.

References and primary resources

1. NCBI GEO: GSE163857 https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE163857
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10. Love MI, Huber W, Anders S. DESeq2. Genome Biology. 2014;15:550. Current Bioconductor vignette used for count-matrix, design, and interaction guidance. https://bioconductor.org/packages/release/bioc/vignettes/DESeq2/inst/doc/DESeq2.html
11. biomaRt Bioconductor vignette: accessing Ensembl annotation and homolog information. https://bioconductor.org/packages/release/bioc/vignettes/biomaRt/inst/doc/accessing_ensembl.html

Scope of this tutorial: This guide explains a defensible analysis plan and provides adaptable code templates. It does not report newly calculated biological results. Public metadata and software can change; re-check GEO sample pages, package documentation, and the original publications when running the analysis.

The one-sentence strategy

Test the complete factor pattern in GSE163857, reproduce the APOE-by-sex signal at single-cell resolution within amyloid in GSE241553, locate the program across cortical cell types with GSE185063, and add APOE2/age evidence from GSE225503 - then compare effect directions without merging studies.

Mouse first. Metadata first. Frozen hypothesis first.